

Ski Safety GPS
Milestone 1 - Detailed Design

Group 9:

Cong Gu (301201228)

Yubin Jeong (301291041)

Tristan Lai (301284657)

ETEC 326 (Sec.001): Technical Project

Professor Mark Thomas

February 9, 2025

Scope Definition

The project aims to design, develop, and deliver a wearable ski safety wristband that enhances skiers' safety by detecting crashes, pinpointing locations, and providing timely alerts to ski resort safety persons. By integrating LoRaWAN communication and GPS tracking in a robust, user-friendly device, this project will address the business need to improve skier safety while reducing the reliance for on-site staff:

Business Objectives:

- **Improve Ski Slope Safety:** Decrease the response times to accidents and enhance overall skier safety standards.
- **Reduce Operation Costs:** Lower labor costs by minimizing the need for additional on-site staff.
- **Enhance Customer Satisfaction:** Build a trustable device that offers reliable safety measures and reassures the user of prompt assistance in cases of emergency.
- **Completed Project:** Launch a functional prototype by the end of the project due date.

Project Deliverables:

- **Wearable Wristband:** A fully functional, weather-resistant, and compact device integrating LoRaWAN module
- **Crash/Inactivity Detection:** Embedded software capable of detecting collisions and extended inactivity
- **Manual Alert:** A user-friendly interface that sends distress signals

Project Exclusions:

- Full smartphone application
- Advanced fitness tracking
- Global satellite communication
- Comprehensive medical response systems

Constraints:

- **Time and Schedules:** Must adhere to the planned timeline with project completion by the course due date
- **Cost:** Each device must be produced at or under \$100 CAD to maintain cost-effective pricing
- **Environmental Conditioning:** Device must function reliably at extreme temperatures
- **Scope and Resources:** Limited engineering and development resources.

Assumptions:

- Reliable LoRaWAN Infrastructure: Ski resorts have or will implement sufficient LoRaWAN gateways to enable communication with the wristband
- User Adoption and Feedback: Users and resort staff will provide timely feedback for project refinement of features and performance
- Industry Standards: All relevant safety, electronics, and communications standards can be met.

In-Scope Components:

1. Hardware and Core Function
 - Wristband device development: Design and prototype a compact, weather-resistant wearable device that is easily attachable to the user's wrist.
 - LoRaWAN technology integration: Utilize LoRaWAN modules and gateways for low-power, long-range communication between the wristband and the end device monitoring computer.
 - Crash and inactivity detection algorithms: Develop accelerometer-based program to recognize sudden impacts or extended motionlessness, triggering the automatic alert. Fine tune false alerts where most alerts are certain.
 - Emergency communication protocols: Install a button that allows skiers to manually send a distress signal.
2. Durability
 - Robust physical design (water-resistant, impact-resistant): Resistant up to IP56 rating.
 - Operational in extreme temperatures (-40°C to +85°C)

Components Acquired (resourcing):

- STM32F103C8T6 – Blue Pill
- GY-NEO6MV2 - GPS Module with Battery
- Centennial Designed Board with RAK chip
- LPS8N Indoor Lora WAN Gateway 915 MHZ

Components on order:

- Plastic Wrist Band
- Casing
- TX RX Signal 4 Pin USB to TTL Adaptor
- Push-button
- SFM-27 DC 3-24V Electronic Buzzer

Scheduling:

Date	Goal
February 16 th ,2025	Hardware pre-assembly, software development (code testing, bootloader & more)
February 23 rd ,2025	Demo Project Assembly and pre-testing with LoRa Wan (Bluetooth implementation)
March 9 th , 2025	Milestone 2 - Preliminary Demo
March 16 th ,2025	<u>Project Testing and improvement</u>
March 23 rd , 2025	Project Testing (scenario implementation: extreme weather, physical contact etc.) and Report
March 30 th ,2025	Final Work Group-assessment and Improvement
April 5 th , 2025	Project Description and Social Innovation (Final Report)
April 6 th , 2025	Working Project Demonstration

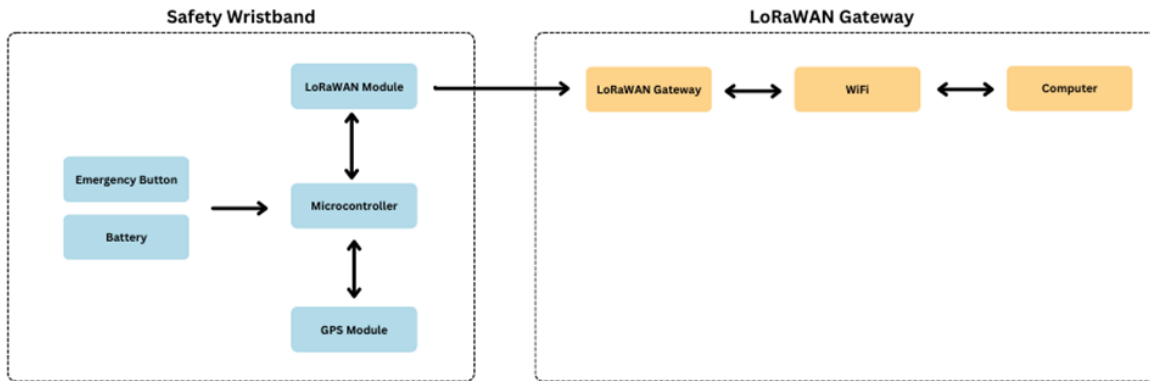
Technology used:

Parts	Descriptions:
RAK-chip	Custom PCB board from Centennial College with a RAK-chip - this chip Enables long-range, low-power communication between the wristband and the LoRaWAN gateway.
Blue Pill	STM32F103C8T6 (Blue Pill) processes data from sensors (GPS, accelerometer) and controls the RAK chip for data transmission.
GY-NEO6MV2 - GPS Module	Sensor that Tracks the skier's location

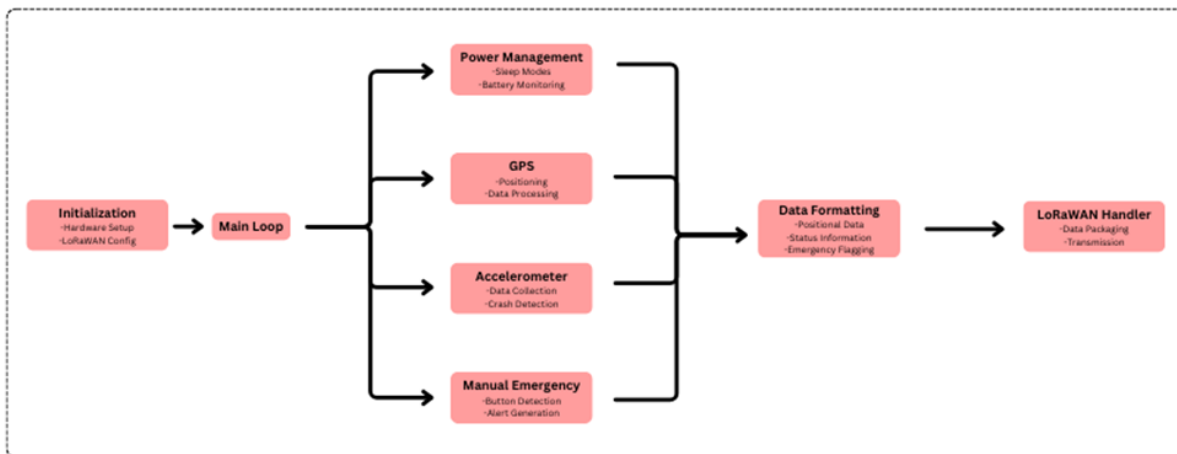
Accelerometer:	Sensor that detects falls or sudden movements.
LPS8N Indoor LoRaWAN Gateway	Bridge between the wristband and the ski-safety-monitoring center. The Gateway receives data from the wristbands and forwards it to the backend system for monitoring and analysis while maintaining long-rang, low-power that also can support multiple wristbands simultaneously
SFM-27 DC 3-24V Electronic Buzzer	Buzzer that produces 100dB at 12VDC

System Block Diagram

Hardware Block Diagram



Software Block Diagram



System Design Notes

Hardware/Software	Function	Description
Button	Emergency Message	When the controllers detects the button press via an interrupt. The system prepares an emergency message which contains the device ID, emergency flag, latitude, and longitude. The data is sent through a packet to the LoRaWAN module which is then sent to the gateway.
Accelerometer	Crash Detection	The accelerometer detects a sudden change in motion. Using an algorithm (e.g. check for prolonged inactivity after the impact). Prepare a packet with the device ID, emergency flag, latitude, and longitude. Sent the packet via the LoRaWAN module to the gateway.
GPS Module	GPS Tracking	The GPS module sends latitude and longitude data to the controller. The latest location is stored within memory for emergency alerts. Sends updates periodically to the LoRaWAN gateway.
Battery/Processor	Power Management	Optimizes battery life by managing the power usage. Puts the GPS and LoRaWAN modules to sleep so it is not constantly pulling. Monitor battery percentage. When battery reaches a certain percentage, the LED will blink.